

VU TRONG CHAU

Manor, TX | +1 (904) 820-9298 | chautrongvu@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

WORK EXPERIENCE

TechX | AI Systems & LLM Engineering Intern

May 2026 - Present

- Built a **multi-LLM inference orchestration pipeline** integrating Claude 3.5, GPT-4, Gemini 1.5 Pro, and Llama 3 with secure API orchestration, retry logic, and rate limiting
- Implemented memory-efficient inference using **4-bit quantization**, FP16, and CPU offloading with **Hugging Face Transformers**
- Deployed **GPU-accelerated inference services** on **GCP Vertex AI** with Kubernetes-managed autoscaling, batch inference queuing, and response caching

PROJECTS

Healthcare Chatbot | Python, LangGraph, Langchain, Flask, FAISS, Groq, AWS, Docker, RAGAS | [Link](#)

May 2026

- Built a 5-agent LangGraph RAG pipeline over **500K+ medical records**, achieving **90% accuracy** and **0.898 macro F1** with Linear SVM.
- Engineered AWS S3, FAISS vector storage, and ETL pipelines, RAGAS-guided chunking cut unsafe response rate to **<0.3%**
- Deployed Dockerized microservices with GitHub Actions CI/CD, and Prometheus monitoring, pipeline runs end-to-end in under **4 minutes**

Credit Risk Analytics | Python, Tableau, Scikit-learn, Pandas, NumPy, Flask, FRED API | [Link](#)

Dec 2025

- Built end-to-end ML pipelines for loan default prediction using **2.26M+ financial records**, processing **110K+ loans** across **7 Plains states**
- Developed regression and classification models, achieving **R²=0.91** and **AUC of 0.79** with **89.7% accuracy** on imbalanced datasets
- Engineered macroeconomic features and automated ETL pipelines integrating FRED, BLS, and BEA data with zero missing values

Threat Detection using Machine Learning | Python, PyTorch, FastAPI, NLP, Hugging Face, Docker, AWS, CI/CD | [Link](#)

Aug 2025

- Built a production-ready AI threat detection platform using **130K+ labeled texts** to classify threat content in real time using FastAPI, scikit-learn, and Hugging Face Toxic-BERT, achieving **84.9% accuracy** and **0.855 F1-score**
- Developed NLP pipelines, REST APIs, and analytics dashboards using TF-IDF, Logistic Regression, NLTK, PyTorch, Pandas, and SQLite
- Deployed the application with Docker, GitHub Actions CI/CD, AWS EC2/S3, and Hugging Face Spaces for automated cloud deployment

Sleep Quality Prediction | Python, Scikit-learn, Flask, Pandas, NumPy | [Link](#)

Apr 2025

- Built ML pipelines on **110K+ records** to classify sleep disorders and predict sleep quality using multiple machine learning models
- Achieved **82.4% classification accuracy** and **R²=0.67** on regression tasks by engineering behavioral and physiological health features
- Developed a **Flask prediction API** and real-time analytics dashboard with personalized health recommendations and deployed ML models

Global Population Prediction | Python, Scikit-learn, Pandas, D3.js, ETL Pipelines, Flask | [Link](#)

Dec 2024

- Engineered a **ETL pipeline** integrating 12 World Bank indicators across **195+ countries and 60+ years** (1960–2023)
- Selected Linear Regression with the strongest performance with **R²=94.53%**, **MAE=3.11%**, and **MSE=0.60%**
- Built an interactive **D3.js dashboard** with a choropleth map, time-series analytics, and demographic comparison charts

EDUCATION

Troy University

Troy, Alabama

Master of Science in Computer Science - Artificial Intelligence (GPA: 3.5/4.0)

Jul 2025

Coursework: Analysis of Algorithms, Computer Architecture, Machine Learning, Advanced Artificial Intelligence, Data Visualization

Graduate Coursework in Master of Business Administration (MBA) - Business Analytics

May 2026

University of Sunderland

Sunderland, UK

Bachelor of Engineering in Electronic and Electrical Engineering | UK 2:1 Honours

Jul 2021

TECHNICAL SKILLS

Programming Languages: Python, SQL, C/C++, JavaScript

ML / DL Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face, LangChain, LangGraph, Pandas, NumPy, NLTK

Generative AI & LLMs: RAG, Prompt Engineering, FAISS, Chroma, RAGAS, Fine-tuning, Quantization, Agentic Pipelines

MLOps & Cloud: Docker, Kubernetes, CI/CD, GCP Vertex AI, AWS, FastAPI, Triton Inference Server, Tableau, D3.js, ETL Pipelines

CERTIFICATIONS

LLM Application Engineering and Development | **Simplilearn** | [Link](#)

Oct 2025

Data Science Methodology | **IBM** | [Link](#)

Sep 2025

Generative AI with Large Language Models | **DeepLearning.AI** | [Link](#)

Sep 2025